

The Best Open Source Network Intrusion Detection Tools



In enterprises, preventing breaches in the network in order to protect data is a serious matter. Any malware exploit can cost the company a lot. Maintaining networks securely is an aim that all systems administrators hope to achieve. Let us take a look at a few important open source network intrusion detection tools.

In today's world, data breaches, threats, attacks and intrusions are becoming highly sophisticated. Cyber criminals and hackers come up with new methods of gaining access to business and home networks, making a multi-tiered approach to network security an urgent necessity. An Intrusion Detection System (IDS) is, therefore, the most important tool to be deployed to defend the network against the high tech attacks that emerge daily. An IDS, which is a network security tool, is built to detect vulnerability exploits against a target application or computer. It is regarded as a high-end network device or software application that assists the network or systems administrators in monitoring the network or system for all sorts of malicious activities or threats. Any unusual activity is reported to the administrator using a security information and event management (SIEM) system.

There are a wide variety of IDSs available, ranging from antivirus to hierarchical systems, which monitor network traffic. The most common ones are listed below.

- **NIDS:** Network intrusion detection systems are placed at highly strategic points within the network to monitor inbound and outbound traffic from all devices in the network. But scanning all traffic could lead to the creation of bottlenecks, which impacts the overall speed of the network.
- **HIDS:** Host intrusion detection systems run on separate machines or devices in the network, and provide safeguards to the overall network against threats coming from the outside world.
- **Signature based IDS:** Signature based IDS systems monitor all the packets in the network and compare them against the database of signatures, which are pre-configured and pre-determined attack patterns. They work similar to antivirus software.
- **Anomaly based IDS:** This IDS monitors network traffic and compares it against an established baseline. The baseline determines what is considered *normal* for the network in terms of bandwidth, protocols, ports and other devices, and the IDS alerts the administrator against all sorts of unusual activity.
- **Passive IDS:** This IDS system does the simple job of detection and alerting. It just alerts the administrator for any kind of threat and blocks the concerned activity as a preventive measure.